Math 54 — Week 7

Coordinate systems; change of basis; eigenvectors and eigenvalues; the characteristic equation (Lay 4.4, 4.6, 5.1–5.2)

§1 Facts used from Weeks 1–6

(W1) (Week 6) A basis of V is an independent spanning set; every basis of V has $\dim V$ vectors; in a space of dimension n, n independent vectors are a basis, and more than n vectors are dependent.

(W2) (Weeks 4–5) Invertible matrix theorem: for square A, invertible $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ $\det A \neq 0$ $\iff$ the columns form a basis of $\mathbb{R}^n$. For 2×2 matrices, $A^{-1} = \frac{1}{ad-bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

(W3) (Weeks 4–5) $\det(AB) = \det A \det B$, $\det A^T = \det A$, and the determinant of a triangular matrix is the product of its diagonal entries.

(W4) (Weeks 5–6) $\operatorname{Nul} A$ is a subspace, with a basis from the parametric vector form of $A\mathbf{x} = \mathbf{0}$. Row operations do not change the linear relations among the columns of a matrix (the proof of Lay 4.3, Thm. 6).

(W5) (Weeks 3, 5) A linear T is one-to-one if and only if $T(\mathbf{x}) = \mathbf{0}$ only for $\mathbf{x} = \mathbf{0}$; the proof uses only linearity, so it holds for linear maps between any vector spaces.

(W6) (Week 6) If $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is dependent and $\mathbf{v}_1 \neq \mathbf{0}$, some $\mathbf{v}_j$ $(j > 1)$ is a combination of $\mathbf{v}_1, \ldots, \mathbf{v}_{j-1}$ (Lay 4.3, Thm. 4).

§2 This week

§2.1 Coordinate systems (Lay 4.4)

The question. With the basis $\mathbf{b}_1 = (2, 1)$, $\mathbf{b}_2 = (-1, 1)$ of $\mathbb{R}^2$, how do we describe $\mathbf{x} = (4, 5)$ by a list of numbers relative to this basis, why is the list unique, and how do we compute it?

Theorem (The unique representation theorem (Lay 4.4, Thm. 7))
Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for a vector space V. Then for each $\mathbf{x}$ in V there is a unique set of scalars $c_1, \ldots, c_n$ with $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

Proof. 1. $\mathcal{B}$ spans V, so there is at least one such list.

2. If also $\mathbf{x} = d_1\mathbf{b}_1 + \cdots + d_n\mathbf{b}_n$, subtracting gives $\mathbf{0} = (c_1 - d_1)\mathbf{b}_1 + \cdots + (c_n - d_n)\mathbf{b}_n$. $\mathcal{B}$ is independent, so $c_i - d_i = 0$ for every i.

□

Definition. These weights are the **coordinates of $\mathbf{x}$ relative to $\mathcal{B}$**, and $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n)$ in $\mathbb{R}^n$ is the **$\mathcal{B}$-coordinate vector** of $\mathbf{x}$. The map $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is the **coordinate mapping** determined by $\mathcal{B}$.

Coordinates in $\mathbb{R}^n$. Let $P_{\mathcal{B}} = [\,\mathbf{b}_1 \cdots \mathbf{b}_n\,]$, the **change-of-coordinates matrix** from $\mathcal{B}$ to the standard basis. By the definition of a matrix times a vector,

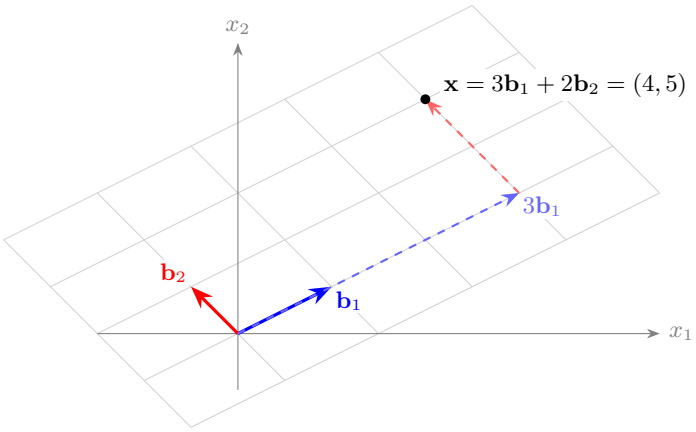
$$\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}, \qquad \text{so} \qquad [\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}.$$

$P_{\mathcal{B}}$ is invertible because its columns are a basis of $\mathbb{R}^n$ (W2). In practice, find $[\mathbf{x}]_{\mathcal{B}}$ by row reducing $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\; \mathbf{x}\,]$.

The question answered. Solve $c_1\mathbf{b}_1 + c_2\mathbf{b}_2 = \mathbf{x}$; interchange first to avoid fractions:

$$\left[\begin{array}{cc|c} 2 & -1 & 4 \\ 1 & 1 & 5 \end{array}\right] \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & -1 & 4 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & -3 & -6 \end{array}\right] \xrightarrow[R_1 - R_2]{-\frac{1}{3}R_2} \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array}\right], \qquad [\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Check: $3(2,1) + 2(-1,1) = (4,5)$. The list is unique by Theorem 7.



The grid of the basis $\mathcal{B}$: from $\mathbf{0}$ go 3 steps along $\mathbf{b}_1$, then 2 steps along $\mathbf{b}_2$. In standard coordinates the same point is $(4,5)$.

Theorem (The coordinate mapping (Lay 4.4, Thm. 8))
Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ be a basis for V. Then $\mathbf{x} \mapsto [\mathbf{x}]_{\mathcal{B}}$ is a one-to-one linear transformation from V onto $\mathbb{R}^n$.

Proof. 1. **Linear.** If $\mathbf{u} = \sum c_i\mathbf{b}_i$ and $\mathbf{w} = \sum d_i\mathbf{b}_i$, then $\mathbf{u} + \mathbf{w} = \sum(c_i + d_i)\mathbf{b}_i$ and $r\mathbf{u} = \sum(rc_i)\mathbf{b}_i$. By uniqueness these weights are the coordinates, so $[\,\mathbf{u} + \mathbf{w}\,]_{\mathcal{B}} = [\,\mathbf{u}\,]_{\mathcal{B}} + [\,\mathbf{w}\,]_{\mathcal{B}}$ and $[\,r\mathbf{u}\,]_{\mathcal{B}} = r[\,\mathbf{u}\,]_{\mathcal{B}}$.

2. **One-to-one.** If $[\,\mathbf{x}\,]_{\mathcal{B}} = \mathbf{0}$, then $\mathbf{x} = 0\mathbf{b}_1 + \cdots + 0\mathbf{b}_n = \mathbf{0}$. By (W5) the map is one-to-one.

3. **Onto.** Each $(c_1, \ldots, c_n)$ in $\mathbb{R}^n$ is the coordinate vector of $c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$.

□

A one-to-one linear map from V onto W is an **isomorphism**. By Theorem 8, every vector space with a basis of n vectors is isomorphic to $\mathbb{R}^n$.

Coordinates keep linear relations. For $\mathbf{u}_1, \ldots, \mathbf{u}_p$ in V and scalars c_j, linearity of the coordinate map (Theorem 8) gives

$$c_1[\,\mathbf{u}_1\,]_{\mathcal{B}} + \cdots + c_p[\,\mathbf{u}_p\,]_{\mathcal{B}} = [\,c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p\,]_{\mathcal{B}},$$

and a coordinate vector is $\mathbf{0}$ only for the vector $\mathbf{0}$. So

$$c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p = \mathbf{0} \quad \Longleftrightarrow \quad c_1[\,\mathbf{u}_1\,]_{\mathcal{B}} + \cdots + c_p[\,\mathbf{u}_p\,]_{\mathcal{B}} = \mathbf{0}.$$

Hence $\{\mathbf{u}_j\}$ is independent exactly when $\{[\,\mathbf{u}_j\,]_{\mathcal{B}}\}$ is, and in the same way $\mathbf{y}$ is in $\mathrm{Span}\{\mathbf{u}_j\}$ exactly when $[\,\mathbf{y}\,]_{\mathcal{B}}$ is in $\mathrm{Span}\{[\,\mathbf{u}_j\,]_{\mathcal{B}}\}$. Span and independence questions in V become the same questions about columns of numbers.

Worked example: independence of polynomials. Are $p_1 = 1+2t^2$, $p_2 = 4+t+5t^2$, $p_3 = 3+2t$ independent in $\mathbb{P}_2$? With the standard basis $\{1, t, t^2\}$ their coordinate vectors are $(1,0,2)$, $(4,1,5)$, $(3,2,0)$. Row reduce the matrix with these columns:

$$\begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 2 & 5 & 0 \end{bmatrix} \xrightarrow{R_3-2R_1} \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 0 & -3 & -6 \end{bmatrix} \xrightarrow{R_3+3R_2} \begin{bmatrix} 1 & 4 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1-4R_2} \begin{bmatrix} 1 & 0 & -5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Column 3 of the reduced form is -5 times column 1 plus 2 times column 2. Row operations keep column relations (W4), and coordinates keep relations, so

$$p_3 = -5p_1 + 2p_2 : \qquad -5(1+2t^2) + 2(4+t+5t^2) = 3+2t.$$

The polynomials are dependent.

Worked example: coordinates on a subspace. Let $\mathbf{v}_1 = (1,2,0)$, $\mathbf{v}_2 = (0,1,1)$, $H = \mathrm{Span}\{\mathbf{v}_1, \mathbf{v}_2\}$ (a plane in $\mathbb{R}^3$), and $\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$, a basis of H (two vectors, not multiples). Is $\mathbf{x} = (2,7,3)$ in H, and if so, what is $[\mathbf{x}]_{\mathcal{B}}$?

$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 2 & 1 & 7 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{R_2-2R_1} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 1 & 3 \end{array}\right] \xrightarrow{R_3-R_2} \left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & 3 \\ 0 & 0 & 0 \end{array}\right]$$

The system is consistent, so $\mathbf{x} \in H$, and $[\mathbf{x}]_{\mathcal{B}} = (2,3)$: a vector in $\mathbb{R}^2$, although $\mathbf{x}$ is in $\mathbb{R}^3$. Check: $2(1,2,0) + 3(0,1,1) = (2,7,3)$.

Shortcuts

- **Coordinates in $\mathbb{R}^n$**: row reduce $[\,\mathbf{b}_1 \cdots \mathbf{b}_n \;\; \mathbf{x}\,]$. Reason: $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$. Instance: the question.
- **Questions in $\mathbb{P}_n$ or matrix spaces**: pass to coordinates in the standard basis and work with columns of numbers. Reason: coordinates keep linear relations. Instance: the polynomial example.
- **Coordinates of a polynomial in a non-standard basis**: match the coefficients of $1, t, t^2$ to get a linear system for the weights. Reason: two polynomials are equal exactly when their coefficients are equal. Instance: in $\mathbb{P}_1$ with $\mathcal{B} = \{1+t,\ 1-t\}$, $c_1(1+t) + c_2(1-t) = 5+t$ gives

  $$c_1 + c_2 = 5, \qquad c_1 - c_2 = 1,$$

  so $c_1 = 3$, $c_2 = 2$ and $[5+t]_{\mathcal{B}} = (3,2)$.

Traps

- The order of the basis matters: with $\mathcal{B}' = \{\mathbf{b}_2, \mathbf{b}_1\}$, the same $\mathbf{x} = (4,5)$ has $[\mathbf{x}]_{\mathcal{B}'} = (2,3)$, not $(3,2)$.
- List polynomial coefficients in the order of the basis $\{1, t, t^2\}$: $1+2t^2$ has coordinates $(1,0,2)$, not $(2,0,1)$.
- $P_{\mathcal{B}}$ turns $\mathcal{B}$-coordinates into standard coordinates, not the reverse: in the question $P_{\mathcal{B}}(4,5) = (3,9)$, which is not $[\mathbf{x}]_{\mathcal{B}} = (3,2)$. The correct formula is $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$.
- For a basis of a subspace H, $[\mathbf{x}]_{\mathcal{B}}$ has $\dim H$ entries, not n: in the subspace example it is in $\mathbb{R}^2$.

§2.2 Change of basis (Lay 4.6)

The question. Let $\mathcal{B} = \{(1,1),(1,2)\}$ and $\mathcal{C} = \{(1,0),(2,1)\}$. A vector has $\mathcal{B}$-coordinates $(2,-1)$. How do we find its $\mathcal{C}$-coordinates directly, without converting to standard coordinates first?

Theorem (Change of coordinates (Lay 4.6, Thm. 15))

Let $\mathcal{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_n\}$ and $\mathcal{C} = \{\mathbf{c}_1, \ldots, \mathbf{c}_n\}$ be bases of V. Then there is a unique $n \times n$ matrix $P_{\mathcal{C}\leftarrow\mathcal{B}}$ with $[\mathbf{x}]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$ for every $\mathbf{x}$ in V. Its columns are the $\mathcal{C}$-coordinate vectors of the vectors of $\mathcal{B}$:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix}[\mathbf{b}_1]_{\mathcal{C}} & [\mathbf{b}_2]_{\mathcal{C}} & \cdots & [\mathbf{b}_n]_{\mathcal{C}}\end{bmatrix}.$$

It is invertible, and $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = P_{\mathcal{B}\leftarrow\mathcal{C}}$.

The idea

Taking $\mathcal{C}$-coordinates is linear, so it is enough to know the $\mathcal{C}$-coordinates of the vectors $\mathbf{b}_j$: those of any $\mathbf{x} = \sum c_j\mathbf{b}_j$ are the same combination. The *old* basis vectors written in the *new* basis are the columns.

Proof. 1. Write $\mathbf{x} = c_1\mathbf{b}_1 + \cdots + c_n\mathbf{b}_n$, so $[\mathbf{x}]_{\mathcal{B}} = (c_1, \ldots, c_n)$. The $\mathcal{C}$-coordinate map is linear (Theorem 8), so

$$[\mathbf{x}]_{\mathcal{C}} = c_1[\mathbf{b}_1]_{\mathcal{C}} + \cdots + c_n[\mathbf{b}_n]_{\mathcal{C}} = P_{\mathcal{C}\leftarrow\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}.$$

2. Uniqueness: if a matrix M also works, apply both to $[\mathbf{b}_j]_{\mathcal{B}} = \mathbf{e}_j$: column j of M is $M\mathbf{e}_j = [\mathbf{b}_j]_{\mathcal{C}}$.
3. Invertible: if $\sum_j d_j[\mathbf{b}_j]_{\mathcal{C}} = \mathbf{0}$, then $[\sum d_j\mathbf{b}_j]_{\mathcal{C}} = \mathbf{0}$, so $\sum d_j\mathbf{b}_j = \mathbf{0}$ and every $d_j = 0$ since $\mathcal{B}$ is independent. So the columns are independent and $P_{\mathcal{C}\leftarrow\mathcal{B}}$ is invertible (W2).
4. Multiplying step 1 by the inverse gives $[\mathbf{x}]_{\mathcal{B}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}[\mathbf{x}]_{\mathcal{C}}$ for all $\mathbf{x}$. Apply this to $\mathbf{x} = \mathbf{c}_j$, whose $\mathcal{C}$-coordinate vector is $\mathbf{e}_j$: column j of $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}$ is $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}\mathbf{e}_j = [\mathbf{c}_j]_{\mathcal{B}}$, which is column j of $P_{\mathcal{B}\leftarrow\mathcal{C}}$. So $(P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = P_{\mathcal{B}\leftarrow\mathcal{C}}$.

□

Computing $P_{\mathcal{C}\leftarrow\mathcal{B}}$ in $\mathbb{R}^n$. Column j, $[\mathbf{b}_j]_{\mathcal{C}}$, solves $[\mathbf{c}_1 \cdots \mathbf{c}_n]\mathbf{y} = \mathbf{b}_j$. Solve all of them at once:

$$[\mathbf{c}_1 \ \cdots \ \mathbf{c}_n \mid \mathbf{b}_1 \ \cdots \ \mathbf{b}_n] \longrightarrow [I \mid P_{\mathcal{C}\leftarrow\mathcal{B}}].$$

The question answered. With $\mathcal{B} = \{(1,1),(1,2)\}$ and $\mathcal{C} = \{(1,0),(2,1)\}$:

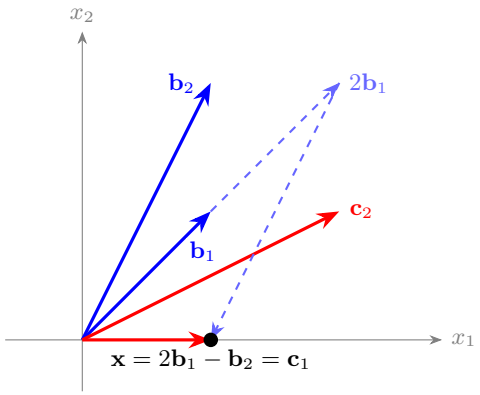
$$\left[\begin{array}{cc|cc} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \end{array}\right] \xrightarrow{R_1 - 2R_2} \left[\begin{array}{cc|cc} 1 & 0 & -1 & -3 \\ 0 & 1 & 1 & 2 \end{array}\right], \qquad P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} -1 & -3 \\ 1 & 2 \end{bmatrix}.$$

Check the first column: $-1(1,0) + 1(2,1) = (1,1) = \mathbf{b}_1$. Then

$$[\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} -1 & -3 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} -2+3 \\ 2-2 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Check: $\mathbf{x} = 2(1,1) - (1,2) = (1,0)$, which is $1\cdot(1,0) + 0\cdot(2,1)$. The other direction uses the 2×2 inverse formula (determinant $-2+3=1$):

$$P_{\mathcal{B}\leftarrow\mathcal{C}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1} = \begin{bmatrix} 2 & 3 \\ -1 & -1 \end{bmatrix}.$$



The question's vector: $[\mathbf{x}]_{\mathcal{B}} = (2,-1)$, and $\mathbf{x} = (1,0)$ is $\mathbf{c}_1$ itself, so $[\mathbf{x}]_{\mathcal{C}} = (1,0)$.

- With $\mathcal{C}$ the standard basis $\mathcal{E}$, $P_{\mathcal{E}\leftarrow\mathcal{B}} = P_{\mathcal{B}}$, because $[\mathbf{b}_j]_{\mathcal{E}} = \mathbf{b}_j$.
- In $\mathbb{R}^n$, $P_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{C}}^{-1}P_{\mathcal{B}}$: go from $\mathcal{B}$ to standard coordinates, then from standard to $\mathcal{C}$.

Worked example: abstract bases. In some space V, $\mathbf{b}_1 = 2\mathbf{c}_1 - \mathbf{c}_2$ and $\mathbf{b}_2 = \mathbf{c}_1 + 3\mathbf{c}_2$. For $\mathbf{x} = 3\mathbf{b}_1 + \mathbf{b}_2$,

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \qquad [\mathbf{x}]_{\mathcal{C}} = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 7 \\ 0 \end{bmatrix}.$$

Check: $3(2\mathbf{c}_1 - \mathbf{c}_2) + (\mathbf{c}_1 + 3\mathbf{c}_2) = 7\mathbf{c}_1$.

Worked example: change of basis in $\mathbb{P}_2$. Let $\mathcal{B} = \{1,\ 1+t,\ 1+t+t^2\}$ and $\mathcal{C} = \{1, t, t^2\}$. The columns of $P_{\mathcal{C}\leftarrow\mathcal{B}}$ are the coefficient vectors of the $\mathcal{B}$ polynomials, and its inverse can be checked by multiplying:

$$P_{\mathcal{C}\leftarrow\mathcal{B}} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}, \qquad P_{\mathcal{B}\leftarrow\mathcal{C}} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

For $p = 2 + 3t + 4t^2$,

$$[p]_{\mathcal{B}} = P_{\mathcal{B}\leftarrow\mathcal{C}} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2-3 \\ 3-4 \\ 4 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 4 \end{bmatrix}.$$

Check: $-1 - (1+t) + 4(1+t+t^2) = 2 + 3t + 4t^2$.

Shortcuts

- **Change of basis in $\mathbb{R}^n$**: row reduce $[\,\mathcal{C} \mid \mathcal{B}\,]$ to $[\,I \mid P_{\mathcal{C}\leftarrow\mathcal{B}}\,]$. Reason: it solves the n systems $[\,\mathbf{c}_1 \cdots \mathbf{c}_n\,]\mathbf{y} = \mathbf{b}_j$ at once. Instance: the question.
- **The other direction**: $P_{\mathcal{B}\leftarrow\mathcal{C}} = (P_{\mathcal{C}\leftarrow\mathcal{B}})^{-1}$; for 2×2 use the inverse formula. Reason: Theorem 15. Instance: the question.
- **Bases given in terms of each other**: the coefficients of $\mathbf{b}_j$ in terms of $\mathcal{C}$ form column j. Reason: Theorem 15. Instance: the abstract example.

Traps

- The order matters: $[\,\mathcal{B} \mid \mathcal{C}\,]$ produces $P_{\mathcal{B}\leftarrow\mathcal{C}}$, the inverse of the matrix wanted. For the question,

$$\left[\begin{array}{cc|cc} 1 & 1 & 1 & 2 \\ 1 & 2 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & -1 \end{array}\right],$$

  which is $P_{\mathcal{B}\leftarrow\mathcal{C}}$, not $P_{\mathcal{C}\leftarrow\mathcal{B}}$.
- In the abstract example, the coefficients of $\mathbf{b}_1$ go into a *column*: writing them as row 1 gives the transpose, which sends $(3,1)$ to $(5,6)$, not $(7,0)$.

§2.3 Eigenvectors and eigenvalues (Lay 5.1)

The question. The matrix below sends $(3,-2)$ to $(-9,11)$, off the line through $(3,-2)$, but it sends $(6,-5)$ to $(-24,20) = -4(6,-5)$, on its own line. Which vectors does a matrix only stretch, and how do they help to compute $A^k\mathbf{x}$?

$$A = \begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix}$$

Definition. An **eigenvector** of an $n \times n$ matrix A is a *nonzero* vector $\mathbf{x}$ with $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is an **eigenvalue** of A if $A\mathbf{x} = \lambda\mathbf{x}$ has a nontrivial solution; such an $\mathbf{x}$ is an eigenvector **corresponding to** λ. The set of all solutions of $(A - \lambda I)\mathbf{x} = \mathbf{0}$ is the **eigenspace** of A for λ.

- λ **is an eigenvalue** $\iff$ $(A-\lambda I)\mathbf{x}=\mathbf{0}$ **has a nontrivial solution**, because $A\mathbf{x}=\lambda\mathbf{x}$ is the same equation as $A\mathbf{x}-\lambda I\mathbf{x}=\mathbf{0}$.
- The eigenspace is $\mathrm{Nul}(A-\lambda I)$, a subspace: $\mathbf{0}$ together with all eigenvectors for λ.
- **Checking a vector**: compute $A\mathbf{u}$ and see whether it is a multiple of $\mathbf{u}$. **Checking a number**: row reduce $A-\lambda I$ and look for a free variable.

Worked example: an eigenspace. Is $\lambda=2$ an eigenvalue of the matrix below, and what is its eigenspace?

$$A=\begin{bmatrix}4&-1&6\\2&1&6\\2&-1&8\end{bmatrix},\qquad A-2I=\begin{bmatrix}2&-1&6\\2&-1&6\\2&-1&6\end{bmatrix}\xrightarrow[R_3-R_1]{R_2-R_1}\begin{bmatrix}2&-1&6\\0&0&0\\0&0&0\end{bmatrix}.$$

1. Two free variables, so $(A-2I)\mathbf{x}=\mathbf{0}$ has nontrivial solutions and 2 is an eigenvalue.
2. The equation is $x_1=\frac{1}{2}x_2-3x_3$, so
$$\mathbf{x}=x_2\begin{bmatrix}1/2\\1\\0\end{bmatrix}+x_3\begin{bmatrix}-3\\0\\1\end{bmatrix},$$
and a basis of the eigenspace is $\{(1,2,0),(-3,0,1)\}$ (the first vector doubled).
3. A multiplies every vector of this plane by 2.

Theorem (Triangular matrices (Lay 5.1, Thm. 1))
The eigenvalues of a triangular matrix are the entries on its main diagonal.

Proof. 1. If A is triangular, so is $A-\lambda I$, with diagonal entries $a_{11}-\lambda,\dots,a_{nn}-\lambda$.

2. λ is an eigenvalue exactly when $(A-\lambda I)\mathbf{x}=\mathbf{0}$ has a nontrivial solution, that is, when $A-\lambda I$ is not invertible, that is, when $\det(A-\lambda I)=0$ (W2).

3. By (W3), $\det(A-\lambda I)=(a_{11}-\lambda)\cdots(a_{nn}-\lambda)$, which is 0 exactly when $\lambda=a_{ii}$ for some i.

□

0 **is an eigenvalue of** A $\iff$ A **is not invertible**, because $A\mathbf{x}=0\mathbf{x}$ is $A\mathbf{x}=\mathbf{0}$.

Theorem (Distinct eigenvalues (Lay 5.1, Thm. 2))
If $\mathbf{v}_1,\dots,\mathbf{v}_r$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1,\dots,\lambda_r$ of an $n\times n$ matrix A, then $\{\mathbf{v}_1,\dots,\mathbf{v}_r\}$ is linearly independent.

The idea

For two vectors: if $\mathbf{v}_2=c\mathbf{v}_1$, applying A gives $\lambda_2\mathbf{v}_2=c\lambda_1\mathbf{v}_1$, while multiplying by λ_2 gives $\lambda_2\mathbf{v}_2=c\lambda_2\mathbf{v}_1$. Subtracting, $c(\lambda_1-\lambda_2)\mathbf{v}_1=\mathbf{0}$, so $c=0$ and $\mathbf{v}_2=\mathbf{0}$, which is impossible. The same subtraction, applied to the first vector that is a combination of earlier ones, gives the general case.

Proof. 1. Suppose the set is dependent. $\mathbf{v}_1\neq\mathbf{0}$, so by (W6) some $\mathbf{v}_j$ is a combination of the vectors before it; let $\mathbf{v}_{p+1}$ be the first such vector ($p\geq 1$). Then $\{\mathbf{v}_1,\dots,\mathbf{v}_p\}$ is independent, since otherwise by (W6) an earlier vector would already be a combination of the ones before it. Write

$$\mathbf{v}_{p+1}=c_1\mathbf{v}_1+\cdots+c_p\mathbf{v}_p.$$

2. Multiply by A, using $A\mathbf{v}_i=\lambda_i\mathbf{v}_i$:
$$\lambda_{p+1}\mathbf{v}_{p+1}=c_1\lambda_1\mathbf{v}_1+\cdots+c_p\lambda_p\mathbf{v}_p.$$

3. Multiply step 1 by λ_{p+1}:
$$\lambda_{p+1}\mathbf{v}_{p+1}=c_1\lambda_{p+1}\mathbf{v}_1+\cdots+c_p\lambda_{p+1}\mathbf{v}_p.$$

4. Subtract: $\mathbf{0} = c_1(\lambda_1 - \lambda_{p+1})\mathbf{v}_1 + \cdots + c_p(\lambda_p - \lambda_{p+1})\mathbf{v}_p$. By independence each $c_i(\lambda_i - \lambda_{p+1}) = 0$, and $\lambda_i \neq \lambda_{p+1}$, so each $c_i = 0$.
5. Then $\mathbf{v}_{p+1} = \mathbf{0}$, which an eigenvector is not. So the set is independent.

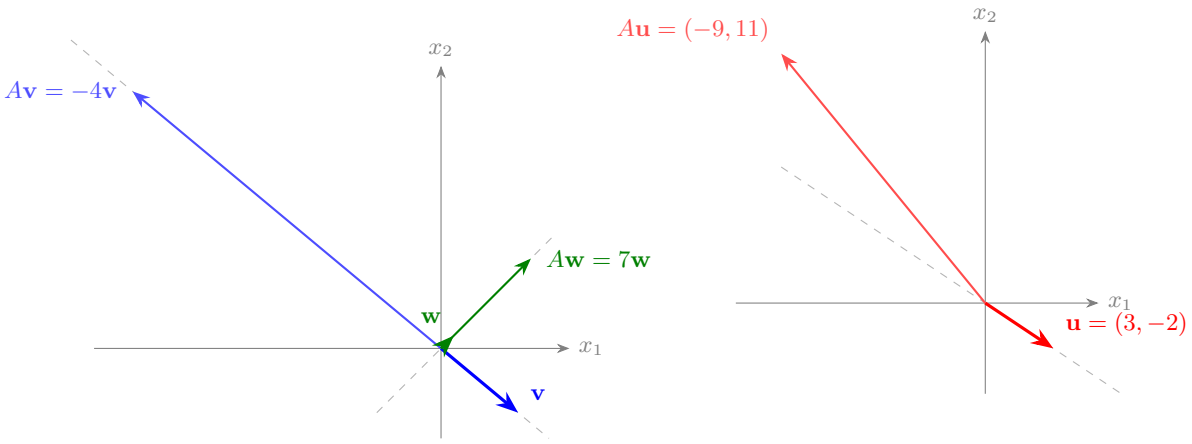
□

An $n \times n$ matrix has at most n distinct eigenvalues: eigenvectors for $n+1$ distinct eigenvalues would be $n+1$ independent vectors in $\mathbb{R}^n$, which is impossible (W1).

Eigenvectors and powers. If $A\mathbf{v} = \lambda\mathbf{v}$, then $A^2\mathbf{v} = A(\lambda\mathbf{v}) = \lambda^2\mathbf{v}$, and repeating, $A^k\mathbf{v} = \lambda^k\mathbf{v}$. If $\mathbf{x}_0 = c_1\mathbf{v}_1 + c_2\mathbf{v}_2$ with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ for λ_1, λ_2, then by linearity

$$A^k\mathbf{x}_0 = c_1\lambda_1^k\mathbf{v}_1 + c_2\lambda_2^k\mathbf{v}_2.$$

This describes the sequence $\mathbf{x}_{k+1} = A\mathbf{x}_k$ (a **difference equation**) for all k at once.



The question's matrix. Left: the eigenvectors $\mathbf{v} = (6, -5)$ and $\mathbf{w} = (1, 1)$ stay on their lines (dashed) and are multiplied by -4 and 7. Right, at a larger scale: $\mathbf{u} = (3, -2)$ is not an eigenvector, and $A\mathbf{u}$ leaves the dashed line through $\mathbf{u}$.

The question answered. The vectors that A only stretches are the eigenvectors. Here $A(1, 1) = (1+6,\ 5+2) = (7, 7)$ and $A(6, -5) = -4(6, -5)$, so 7 and -4 are eigenvalues. Since $(7, -4) = (1, 1) + (6, -5)$,

$$A^k \begin{bmatrix} 7 \\ -4 \end{bmatrix} = 7^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + (-4)^k \begin{bmatrix} 6 \\ -5 \end{bmatrix}$$

for every k, with no matrix powers computed.

Shortcuts

- **Is u an eigenvector?** One multiplication: compare $A\mathbf{u}$ with $\mathbf{u}$. **Is λ an eigenvalue?** One row reduction of $A - \lambda I$, looking for a free variable. Reason: the first item of 5.1. Instance: the question, and $\lambda = 2$ above.
- **Triangular**: the eigenvalues are the diagonal entries; no determinant needed. Reason: Theorem 1. Instance: the matrix below has eigenvalues 2 and -1.

$$\begin{bmatrix} 2 & 5 \\ 0 & -1 \end{bmatrix}$$

- **Equal row sums**: if every row of A adds up to s, then s is an eigenvalue with eigenvector $(1, \ldots, 1)$. Reason: entry i of $A(1, \ldots, 1)$ is the sum of row i. Instance: the question's rows add up to 7.
- **Singular matrices**: 0 is an eigenvalue exactly when A is not invertible. Instance: the matrix below sends $(2, -1)$ to $\mathbf{0}$, so 0 is an eigenvalue.

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

- **Powers and inverses**: if $A\mathbf{x} = \lambda\mathbf{x}$, then $A^k\mathbf{x} = \lambda^k\mathbf{x}$, and if A is invertible, $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$ (multiply $A\mathbf{x} = \lambda\mathbf{x}$ by $\lambda^{-1}A^{-1}$; here $\lambda \neq 0$). Instance: $A^2(6,-5) = 16(6,-5)$.

Traps

- An eigenvector must be nonzero ($A\mathbf{0} = \lambda\mathbf{0}$ for every λ); an eigenvalue may be 0.
- The eigenspace is $\mathrm{Nul}(A - \lambda I)$; row reduce $A - \lambda I$, not A.
- Row reducing A does not preserve its eigenvalues. The question's matrix has eigenvalues 7 and -4, but $R_2 - 5R_1$ turns it into a triangular matrix with eigenvalues 1 and -28:

$$\begin{bmatrix} 1 & 6 \\ 5 & 2 \end{bmatrix} \xrightarrow{R_2 - 5R_1} \begin{bmatrix} 1 & 6 \\ 0 & -28 \end{bmatrix}.$$

- Two independent eigenvectors need not have different eigenvalues: $(1,2,0)$ and $(-3,0,1)$ in the eigenspace example both have eigenvalue 2. Theorem 2 goes only one way.

§2.4 The characteristic equation (Lay 5.2)

The question. Checking one candidate λ at a time is slow. How do we find all eigenvalues of the matrix below at once, together with their eigenvectors?

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Theorem (The characteristic equation (Lay 5.2))
A scalar λ is an eigenvalue of an $n \times n$ matrix A if and only if λ satisfies $\det(A - \lambda I) = 0$.

Proof. 1. $A\mathbf{x} = \lambda\mathbf{x}$ is the same equation as $(A - \lambda I)\mathbf{x} = \mathbf{0}$, so λ is an eigenvalue exactly when this homogeneous system has a nontrivial solution.

2. By the invertible matrix theorem (W2), $(A - \lambda I)\mathbf{x} = \mathbf{0}$ has a nontrivial solution exactly when $A - \lambda I$ is not invertible, which happens exactly when $\det(A - \lambda I) = 0$.

□

For an $n \times n$ matrix, $\det(A - \lambda I)$ is a polynomial in λ of degree n with leading coefficient $(-1)^n$, the **characteristic polynomial** (Lay states this without proof; it is cited here). The **(algebraic) multiplicity** of an eigenvalue is its multiplicity as a root. The **trace** $\operatorname{tr} A$ is the sum of the diagonal entries.

The 2×2 case. Expanding,

$$\det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = (a - \lambda)(d - \lambda) - bc = \lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A.$$

If this has real roots λ_1, λ_2, then it equals $(\lambda - \lambda_1)(\lambda - \lambda_2) = \lambda^2 - (\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2$; comparing coefficients, $\lambda_1 + \lambda_2 = \operatorname{tr} A$ and $\lambda_1\lambda_2 = \det A$. If the discriminant $(\operatorname{tr} A)^2 - 4 \det A$ is negative, there are no real eigenvalues.

The question answered. $\operatorname{tr} A = 6$ and $\det A = 8$, so

$$\det(A - \lambda I) = \lambda^2 - 6\lambda + 8 = (\lambda - 2)(\lambda - 4).$$

The eigenvalues are 2 and 4. For each, $A - \lambda I$ is singular, so its second row is a multiple of its first, and the first row's equation gives the eigenvectors:

$$\lambda = 2: \quad A - 2I = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad x_1 + x_2 = 0, \quad \text{eigenvector } (-1,1);$$

$$\lambda = 4: \quad A - 4I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}, \quad -x_1 + x_2 = 0, \quad \text{eigenvector } (1,1).$$

Check: the eigenvalues add up to $6 = \operatorname{tr} A$ and multiply to $8 = \det A$.

Worked example (3×3). Expand $\det(A - \lambda I)$ down column 1, which has one nonzero entry, and keep the factors:

$$A = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}, \qquad \det(A - \lambda I) = (1-\lambda) \det \begin{bmatrix} 3-\lambda & 0 \\ 1 & 2-\lambda \end{bmatrix} = (1-\lambda)(3-\lambda)(2-\lambda).$$

The eigenvalues are $1, 2, 3$, each of multiplicity one.

More consequences.

- $\det A$ **is the product of the eigenvalues** when the characteristic polynomial factors as $(\lambda_1 - \lambda) \cdots (\lambda_n - \lambda)$ (eigenvalues repeated by multiplicity): set $\lambda = 0$.
- A **and** A^T **have the same eigenvalues**: $\det(A^T - \lambda I) = \det\big((A - \lambda I)^T\big) = \det(A - \lambda I)$ by (W3).
- **The invertible matrix theorem, continued**: A is invertible $\iff$ 0 is not an eigenvalue of A.

Definition. A and B ($n \times n$) are **similar** if $B = P^{-1}AP$ for some invertible P. Changing A into $P^{-1}AP$ is a **similarity transformation**.

Theorem (Similar matrices (Lay 5.2, Thm. 4))
If A and B are similar, they have the same characteristic polynomial, hence the same eigenvalues with the same multiplicities.

Proof. 1. $B - \lambda I = P^{-1}AP - \lambda P^{-1}P = P^{-1}(A - \lambda I)P$.

2. By the product rule (W3),

$$\det(B - \lambda I) = \det(P^{-1}) \det(A - \lambda I) \det P = \det(A - \lambda I),$$

because $\det(P^{-1}) \det P = \det(P^{-1}P) = 1$.

□

Shortcuts

- 2×2: $\det(A - \lambda I) = \lambda^2 - (\operatorname{tr} A)\lambda + \det A$; real eigenvalues sum to the trace and multiply to the determinant. Reason and instance: above.
- 2×2 **eigenvector**: if $b \neq 0$, an eigenvector for λ is $(b,\ \lambda - a)$. Reason: $A - \lambda I$ is singular and its first row $(a - \lambda,\ b)$ is nonzero, so the second row is a multiple of the first, and $(a - \lambda)b + b(\lambda - a) = 0$. Condition: $b \neq 0$. Instance: the question with $\lambda = 4$ gives $(1, 1)$.
- 3×3 **characteristic polynomial**: expand $\det(A - \lambda I)$ along a row or column with zeros, and *keep factors* such as $(1 - \lambda)$ instead of multiplying out. Reason: the roots can then be read off. Instance: the 3×3 example.
- **Check eigenvalues with trace and determinant**. Reason: the comparison of coefficients above. Instance: the question.

Traps

- The equation is $\det(A - \lambda I) = 0$, not $\det A - \lambda = 0$: for the question's matrix, $\det A - \lambda = 8 - \lambda$ would give the wrong eigenvalue 8.
- A real matrix may have no real eigenvalues. The matrix below has $\operatorname{tr} = 4$, $\det = 5$, and $\lambda^2 - 4\lambda + 5$ has discriminant $16 - 20 < 0$; the trace and determinant rule then says nothing about real eigenvalues.

$$\begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}$$

- The same characteristic polynomial does not make matrices similar. The two matrices below both have characteristic polynomial $(2-\lambda)^2$, but $P^{-1}(2I)P = 2I$ for every P, so only $2I$ is similar to $2I$.

$$\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \qquad \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$$

- Similarity is not row equivalence: those same two matrices are row equivalent (both reduce to I), but they are not similar.

§3 Homework this week

HW6, Sections 4.4 and 4.6, and HW7, Sections 5.1–5.2.

Section 4.4.

- **#1, #3** (find $\mathbf{x}$ from $[\mathbf{x}]_{\mathcal{B}}$). $\mathbf{x} = P_{\mathcal{B}}[\mathbf{x}]_{\mathcal{B}}$.
- **#7** (find $[\mathbf{x}]_{\mathcal{B}}$). Row reduce $[\,\mathbf{b}_1\ \mathbf{b}_2\ \mathbf{b}_3\ \ \mathbf{x}\,]$.
- **#13** (coordinates of a polynomial). Match the coefficients of $1, t, t^2$ and solve the system (P2(a)).
- **#15, #17, #19** (true/false). The definition of $[\mathbf{x}]_{\mathcal{B}}$; $[\mathbf{x}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{x}$ (third trap); $\dim \mathbb{P}_3 = 4$.
- **#25** (the matrix of the coordinate map). It is $P_{\mathcal{B}}^{-1}$.
- **#36** (polynomials as a basis; coordinates of q). Coordinate vectors in the standard basis, then the basis theorem; then solve for the weights.

Section 4.6.

- **#1, #5, #6** (abstract bases). The columns are the coordinates of the old basis vectors in the new basis; then multiply (abstract example).
- **#9** (bases of $\mathbb{R}^2$). Row reduce $[\,\mathbf{c}_1\ \mathbf{c}_2 \mid \mathbf{b}_1\ \mathbf{b}_2\,]$; the other matrix is its inverse.
- **#12, #13** (true/false). The columns are coordinate vectors of an independent set; with $\mathcal{C}$ standard, $P_{\mathcal{C}\leftarrow\mathcal{B}} = P_{\mathcal{B}}$.
- **#15** (polynomial bases). Coefficient vectors as columns (the $\mathbb{P}_2$ example).

Section 5.1.

- **#3, #5** (is it an eigenvector/eigenvalue?). Compute $A\mathbf{u}$ and compare with $\mathbf{u}$.
- **#7** (is 4 an eigenvalue?). Row reduce $A - 4I$ and look for a free variable.
- **#9, #13** (eigenspace bases). Parametric vector form of $(A-\lambda I)\mathbf{x} = \mathbf{0}$.
- **#23, #24, #27, #28, #29, #30** (true/false). 0 eigenvalue $\iff$ singular; $A\mathbf{x} = \lambda\mathbf{x}$ is $(A-\lambda I)\mathbf{x} = \mathbf{0}$; the last trap of 5.1; Theorem 1 needs a triangular matrix; $A(2\mathbf{v}) = 2(A\mathbf{v})$; the eigenspace is a null space.
- **#34** ($A^2 = 0$). If $A\mathbf{x} = \lambda\mathbf{x}$ then $\mathbf{0} = A^2\mathbf{x} = \lambda^2\mathbf{x}$.

Section 5.2.

- **#5, #7** (2×2 characteristic polynomials). $\lambda^2 - (\operatorname{tr} A)\lambda + \det A$; for #7 look at the discriminant.
- **#11, #13** (3×3). Expand along a row or column with zeros and keep the factors.
- **#15** (triangular). Read the diagonal, with multiplicities.
- **#18** (choose h). The eigenspace for 5 is $\operatorname{Nul}(A-5I)$; choose h so that $A-5I$ has two free variables.
- **#19** ($\det A$ and the eigenvalues). Set $\lambda = 0$ in the factored characteristic polynomial.
- **#24, #25** (true/false). Theorem 4; the characteristic equation.

§4 Problem set

P1. Find $[\mathbf{x}]_{\mathcal{B}}$ for $\mathbf{x} = (4, 9)$ and $\mathcal{B} = \{(1, 3), (2, 5)\}$.

P2. (a) $\mathcal{B} = \{1 + t,\ t + t^2,\ 1 + t^2\}$ is a basis of $\mathbb{P}_2$. Find $[p]_{\mathcal{B}}$ for $p = 3 + 2t - t^2$. (b) Use coordinate vectors to decide whether $1 - t$, $t - t^2$, $1 - t^2$ are linearly independent.

P3. $\mathcal{B} = \{(1, 2), (3, 4)\}$ and $\mathcal{C} = \{(1, 1), (0, 1)\}$. Find $P_{\mathcal{C} \leftarrow \mathcal{B}}$ and $P_{\mathcal{B} \leftarrow \mathcal{C}}$. If $[\mathbf{x}]_{\mathcal{B}} = (2, -1)$, find $[\mathbf{x}]_{\mathcal{C}}$.

P4. For the matrix below, is $\mathbf{v} = (1, -1, 0)$ an eigenvector? If so, for which eigenvalue? Is 4 an eigenvalue?

$$A = \begin{bmatrix} 3 & 0 & 1 \\ -1 & 2 & -1 \\ 0 & 0 & 1 \end{bmatrix}$$

P5. For the same A, find the characteristic polynomial, all eigenvalues, and a basis of the eigenspace for $\lambda = 1$.

P6. Find the characteristic polynomials and eigenvalues of the matrices below.

$$\begin{bmatrix} 2 & 7 \\ 7 & 2 \end{bmatrix} \qquad \begin{bmatrix} 5 & -3 \\ -4 & 3 \end{bmatrix}$$

P7. List the eigenvalues, with multiplicities, of the matrix below. Is it invertible?

$$\begin{bmatrix} 2 & 3 & -1 & 4 \\ 0 & -1 & 5 & 2 \\ 0 & 0 & 2 & 7 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

P8. Suppose $A\mathbf{x} = \lambda\mathbf{x}$ with $\mathbf{x} \neq \mathbf{0}$. (a) Show that $\mathbf{x}$ is an eigenvector of A^2 and of $A + 3I$; for which eigenvalues? (b) If A is invertible, show $\lambda \neq 0$ and $A^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$.